

AWS Academy Lab 2: Cloud Costing & Network Foundations

A step-by-step hands-on guide for AWS Academy Learner Lab environments — covering financial blueprinting with the AWS Pricing Calculator and deploying live foundational network infrastructure inside the AWS Management Console.

ITNW1009

AWS ACADEMY

HANDS-ON LAB



Lab Environment Profile

Platform & Constraints

This lab runs inside the **AWS Academy Sandbox (Learner Lab)**. Students operate under restricted IAM execution profiles. Service Control Policies (SCPs) block global configuration permissions and live billing tier activations.

 All compute deployments must use **t2.micro** instances and stay within the **\$100 sandbox budget cap**.

Lab Scope Overview

Part A — Financial Blueprint

Build a multi-tier cost estimate inside the **AWS Pricing Calculator**. Model real-world EC2 and EBS configurations without spending real money.

Part B — Live Network Sandbox

Deploy a functional **VPC, Subnet, and Security Group** inside the AWS Console to verify structural mechanics safely within sandbox limits.



PART A

Financial Blueprinting via the AWS Pricing Calculator

The AWS Pricing Calculator is a free, browser-based planning tool that lets you model infrastructure costs before committing any real resources. In Part A, you will estimate the monthly operational expenditure (OpEx) of a 10-server web application tier with attached EBS storage – building habits that mirror real-world cloud financial governance.

Initialize the Estimation Portal

1 Open the AWS Pricing Calculator

In a new browser tab, navigate to **<https://calculator.aws/#/>** — the official AWS cost planning portal. No AWS account login is required to begin.

2 Click "Create Estimate"

Locate and click the prominent primary action button labeled **Create estimate** on the landing page. This initializes a blank project workspace where you will add and configure services.

i The Pricing Calculator always reflects current AWS public pricing. Estimates are shareable via a public link — you will generate one at the end of Part A.

Configure the Region & EC2 Compute Tier

01

Search for Amazon EC2

In the "Find Service" search bar, type **Amazon EC2** and press Enter. Click **Configure** on the EC2 card result.

02

Set Description & Region

Enter **Web Application Server Tier** in the Description field. Set the Region dropdown to **US East (N. Virginia)**. Ensure OS is set to **Linux**.

03

Set Workload & Instance Count

Under Workload, select **Constant Usage**. Set the **Number of instances** to exactly **10**.

04

Select Instance Type & Pricing

Search for and select **t2.micro** (1 vCPU, 1 GiB Memory). Under Pricing Strategy, select **On-Demand** to model true utility operational costs.

Configure Attached EBS Storage Blocks

After configuring your EC2 instances, scroll down on the same configuration screen to the **Amazon Elastic Block Store (EBS)** section. EBS volumes provide persistent block-level storage that remains intact even if an EC2 instance is stopped or restarted.

Volume Type

Set to **General Purpose SSD (gp3)** – the default free-tier baseline for most workloads.

Storage Amount

Input **30 GB** in the Storage amount field. Leave IOPS and throughput toggles at their defaults.

Save Configuration

Click **Save and add to my estimate** at the bottom right corner of the page.



Export the Financial Summary

1

View Monthly Estimate

You are redirected to the **My Estimate** dashboard showing a calculated monthly run rate for your 10 EC2 instances with 30 GB EBS storage each.

2

Click "Share"

Locate the **Share** button at the top right of the screen. A modal pop-up window will appear with sharing options.

3

Copy Public Link

Click **Copy public link** inside the modal. Paste the URL into a local notepad file immediately – this link **must** be submitted in your final lab report.



Do not close the browser tab before saving your public link. Unsaved estimate URLs cannot be recovered from the session.

PART B

Foundational Network Sandbox Implementation

In Part B, you shift from cost modeling to live infrastructure deployment. You will build three foundational AWS networking components — a **VPC**, a **Subnet**, and a **Security Group** — inside the real AWS Management Console, using your AWS Academy Learner Lab sandbox environment. Each step reinforces the structural mechanics of cloud networking in a safe, budget-controlled setting.



Accessing the AWS Academy Sandbox

01

Log Into AWS Academy

Sign in to your university **AWS Academy portal** account and navigate to your active course: **ITNW 1009: Fundamentals of Cloud and AI**.

02

Open the Learner Lab

Click **Modules**, then locate and select **AWS Academy Learner Lab**. On the initialization dashboard, click **Start Lab**.

03

Wait for Green Status

The lab environment takes **2–3 minutes** to spin up. Wait until the status circle changes from **amber** to a solid **green**, indicating the sandbox is active.

04

Open the AWS Console

Click the **AWS** text link immediately to the left of the green status dot. This opens the live AWS Management Console in a secure secondary browser tab.

Building the Isolated Virtual Private Cloud (VPC)

01

Navigate to VPC

In the top global search bar, type **VPC** and select **VPC (Isolated Cloud Resources)**. Click the orange **Create VPC** button on the dashboard.

02

Select "VPC Only"

Under VPC settings, select the **VPC only** radio toggle. Do **not** select "VPC and more" — this violates AWS Academy SCPs.

03

Enter Name & CIDR Block

Name tag: **ITNW1009-Lab2-VPC**

IPv4 CIDR: **10.0.0.0/16** (manual input)

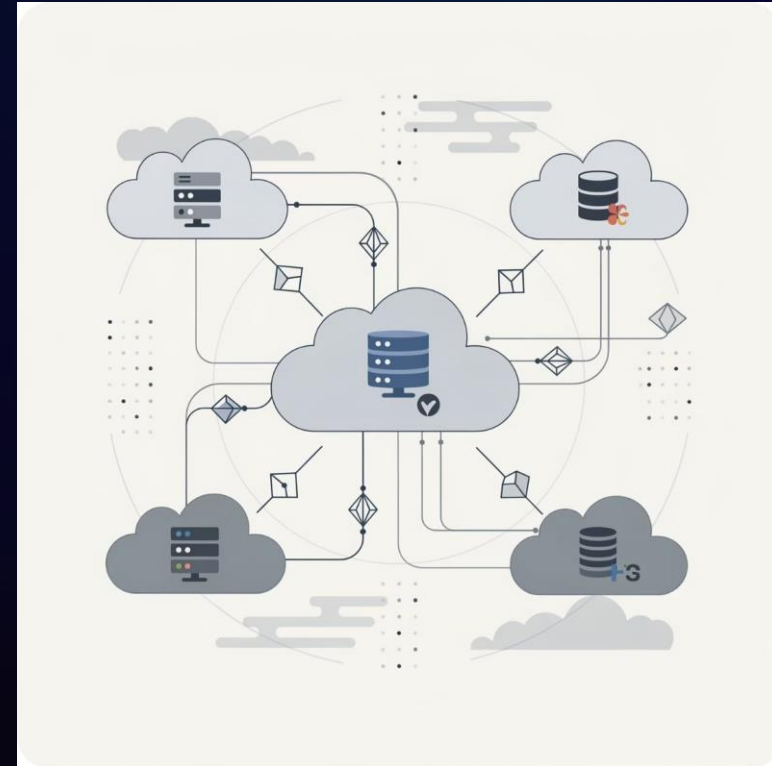
IPv6: **No IPv6 CIDR block**

Tenancy: **Default**

04

Create the VPC

Scroll to the bottom of the form and click **Create VPC**. Confirm the status shows **Available** in the VPC dashboard.



Provisioning an Isolated Testing Subnet

A subnet carves a smaller, addressable block from within your VPC's larger CIDR range. You will create one public-facing subnet inside the N. Virginia Availability Zone **us-east-1a**.

Navigate to Subnets

In the left sidebar, click **Subnets**. Then click **Create subnet** at the top right.

Select Your VPC

In the **VPC ID** dropdown, select **ITNW1009-Lab2-VPC** — your newly created network.

Configure Subnet Details

Subnet name: **Lab2-Public-Subnet-A**

Availability Zone: **us-east-1a**

IPv4 CIDR: **10.0.1.0/24**

Create the Subnet

Click **Create subnet** at the bottom. Verify it appears linked to ITNW1009-Lab2-VPC in the table view.

Engineering the Infrastructure Security Group



01

Navigate to Security Groups

In the left sidebar, scroll to Security and click **Security Groups**. Click **Create security group**.

02

Enter Group Details

Name: **Lab2-WebServer-SG**

Description: **Allow inbound HTTP traffic for testing compute tiers**

VPC: Select **ITNW1009-Lab2-VPC** (clear the default first).

03

Add Inbound Rule

Click **Add rule** in the Inbound rules section.

Type: **HTTP (Port 80)**

Source: **Anywhere-IPv4 (0.0.0.0/0)**

04

Create the Security Group

Click **Create security group** at the bottom. The 0.0.0.0/0 source allows web traffic from any global client IP address.

Lab Validation Checklist

Before closing your console session, verify all three deployed components against the criteria below to earn full lab completion credit.

1

VPC Validation

Navigate to the VPC dashboard.
Confirm **ITNW1009-Lab2-VPC** shows status **Available** with CIDR address **10.0.0.0/16** exactly as configured.

2

Subnet Association

Open the Subnets table view. Confirm **Lab2-Public-Subnet-A** is associated with **ITNW1009-Lab2-VPC** and locked to Availability Zone **us-east-1a**.

3

Security Group Firewall

Open the Security Group details for **Lab2-WebServer-SG**. Confirm an active inbound rule allowing **Port 80 (HTTP)** from source **0.0.0.0/0**.



All three checks passing confirms your sandbox deployment is structurally sound and ready for submission.



Technical Troubleshooting Guide

Two common failure patterns occur in this lab. Understanding the root cause — not just the fix — is an essential cloud engineering skill. Review both symptoms below before starting your lab session.

Symptom A: "Create VPC" Button Greyed Out or IAM Error

Root Cause

The student has either selected a **restricted service region** outside the sandbox's permitted boundaries, or attempted to use the **"VPC and More"** wizard toggle — which calls backend route table configurations that violate AWS Academy Service Control Policies (SCPs).

Resolution Steps

- Check the top-right region dropdown in the AWS Console. It must read **N. Virginia (us-east-1)**. If it reads any other region, switch to N. Virginia to instantly restore sandbox creation rights.
- On the VPC creation form, verify the **VPC Only** radio button is selected — never "VPC and more."

Symptom B: Subnet CIDR Overlap Error

Root Cause

A red warning message reads: *"CIDR Address overlaps with an existing component resource allocation map."* This occurs when the student has entered **typo values** in either the parent VPC CIDR or the subnet CIDR field, causing a conflict with subnet calculation mechanics.

Resolution Steps

- Delete any failed subnet inputs. Return to Step 2 and confirm that the parent VPC CIDR reads exactly **10.0.0.0/16** – a 16-bit mask covering the full private address space.
- Ensure your subnet uses a **24-bit sub-divided range**, typed exactly as **10.0.1.0/24**. Any deviation will trigger the overlap error.

LAB DEBRIEF

Section 3: Academic Review & Debrief

The following discussion prompts are designed to bridge hands-on lab actions to real-world cloud architecture decision-making. Work through each prompt individually or in small groups to reinforce today's core concepts before the session closes.



Connecting the Calculator to Real Networks

"A business manager asks you to move those 10 servers to the AWS Tokyo region (ap-northeast-1) overnight to support a Japanese product launch. What steps do you take inside your Pricing Calculator blueprint, and how will that change your estimated monthly OpEx?"

Calculator Changes

- Open your saved estimate and edit the EC2 configuration
- Change the Region dropdown from **US East (N. Virginia)** to **Asia Pacific (Tokyo)**
- Re-save and generate a new public share link for the updated estimate

Cost Implications

- AWS pricing varies by region — Tokyo on-demand rates for t2.micro are typically **higher** than N. Virginia
- Data transfer costs between regions add additional OpEx
- Always re-estimate before committing to a regional migration

Evaluating the Security Layer

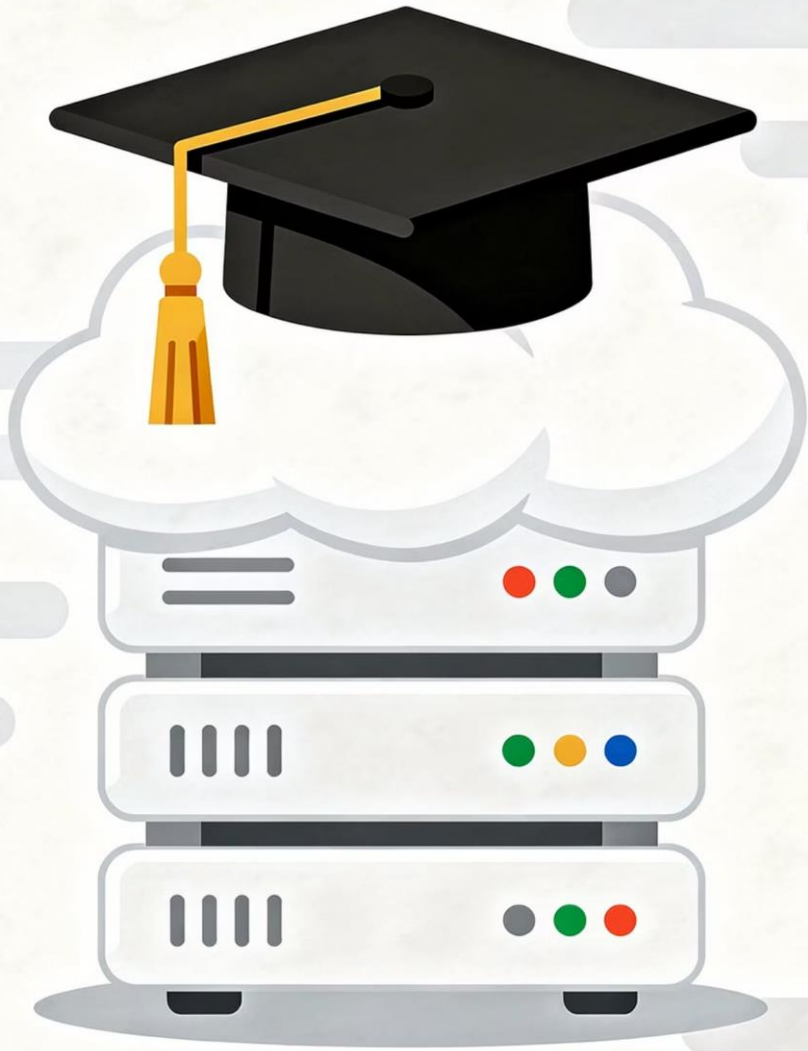
"We opened Port 80 to 0.0.0.0/0. Why is this fine for a public web server – but a dangerous risk for a backend database holding private financial records?"

Public Web Tier (Port 80 Open)

A web server's entire purpose is to respond to browser requests from any client on the internet. Opening Port 80 to **0.0.0.0/0** is intentional and required for public HTTP access. Every legitimate visitor's request arrives via this rule.

Backend Database (Port 80 Open = Critical Risk)

A database server should **never** be reachable from the open internet. It should only accept connections from trusted application servers within the same VPC, using tightly scoped source CIDR rules or security group references. Exposing 0.0.0.0/0 on a database invites brute-force attacks, data exfiltration, and regulatory violations.



Lab Summary: Key Takeaways



Cost Modeling First

Always blueprint your architecture in the **AWS Pricing Calculator** before deploying live infrastructure. Region selection directly impacts monthly OpEx.



VPC Isolates Your Network

A VPC with a /16 CIDR block creates a private, isolated virtual network. Subnets with /24 blocks subdivide it into logical zones tied to physical Availability Zones.



Security Groups = Firewall Rules

Security Groups control inbound and outbound traffic at the instance level. Open only the ports and sources that your workload explicitly requires.



Sandbox Constraints Are Real

AWS Academy SCPs enforce region and service restrictions. Always work in **us-east-1** and use **VPC Only** mode to stay within sandbox permissions.